MILITARY
SPECIFICATION

MIL-R-11047B(SigC)
27 October 1958
Superseding
MIL-C-11047A(SigC)
24 February 1954

Recorder, Code, Tape
(Code Recorder RD-60( )/U)

1. SCOPE

1.1 Scope.- This specification covers one type of equipment designated as Code Recorder RD-60( )/U. This equipment is a portable self-contained unit for recording of telegraph code signals at speeds ranging from 5 to 1000 words per minute. (See 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawing, and publications, of the issue in effect on date of invitation for bids, form a part of this specification.

FEDERAL

QQ-S-571	Solder; lead alloy, tin lead alloy, and tin alloy; flux cored ribbon and wire, and solid form.
T-T-871	Twine; Cotton, wrapping
UU-T-111	Tape; paper gummed (sealing and securing)
LLL-F-291	Fiber board, corrugated, single face (flexible)
PPP-B-566	Boxes, folding, paperboard
PPP-B-621	Boxes, wood, nailed and lock-corner
PPP-B-636	Boxes, fiber
PPP-T-76	Tape; pressure sensitive adhesive, waterproof for packaging and sealing

MILITARY

MIL-E-1	Electron tubes and crystal rectifiers.
MIL-P-75	Packaging, packing, and container-marking of electron tubes
MIL-W-76	Wire and cable, hook-up electrical, insulated
MIL-P-116	Preservation, methods of
MIL-B-117	Bags, interior, packaging
MIL-T-152	Treatment, moisture-and-fungus-resistant, of communications, electronic, and associated electrical equipment
MIL-C-3432	Cable, flexible and extra flexible, 300 and 600 volts.
MIL-W-3861	Wire electrical (bare copper)
MIL-C-3767	Connectors, electrical (power, bladed type)
MIL-C-3885	Cable assemblies and cord assemblies, electrical (for use in electronics, communication and associated electrical equipment).
MIL-B-7883	Brazing of steels, coppers, copper alloys, and nickel alloys.

FSC 6940

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the current version before use.

MIL-W-12349	Wire, hook-up monochlorotrifluoroethylene
MIL-M-13231	Marking of electronic items
MIL-F-14072	Finishes for ground signal equipment
MIL-I-17384	Insulating compound, electrical, quick-drying

STANDARDS

MIL-STD-22	Welded-joint designs
MIL-STD-105	Sampling procedures and tables for inspection by attributes
MIL-STD-109	Inspection terms and definitions
MIL-STD-129	Marking for shipment and storage
MIL-STD-169	Extreme-temperature cycle
MIL-STD-170	Moisture resistance test cycle for ground signal equipment
MIL-STD-200	Electron tubes
MIL-STD-252	Wired equipment, classification of visual and mechanical defects

DRAWINGS

SIGNAL CORPS

SC-D-25540	Code Recorder RD-60( )/U drawing list

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

COMMON CARRIERS

a. For rail

(1) Uniform Freight Classification No. 4

Uniform Freight Classification
202 Union Station
Chicago 6, Illinois

(2) *Consolidated Freight Classification Rules No. 21.*

Consolidated Classification Committee
1 Park Avenue and 33rd Street
New York 16, New York

b. For truck

National Motor Freight Classification No. A-3 and No. 14

Source: https://assist.dla.mil -- Downloaded: 202640-17T15:02Z
Check the source to verify that this is the current version before use.

American Trucking Association Inc.
1424 Sixteenth Street, N.W.
Washington, 6, D.C.

3. REQUIREMENTS

3.1 Description. - Code recorder RD-60( )/U is an undulating type ink-slip recorder with a specialized amplifier which converts code signals to operate the mechanism which applies an inked signal line to a paper tape. The equipment is used to record the signals of an operator using either a semiautomatic or hand telegraph key, to record high-speed radiotelegraph code signals from the output of a commercial type radio receiver with the recorder connected directly to the receiver, or to receive the same type of signals from a radio receiving station over an audio transmission line. Code signals can be recorded at speed up to 1,000 words per minute.

3.2 Construction. - The equipment shall be constructed in accordance with Drawing SC-D-25540. However, if specified performance characteristics cannot be obtained by strict conformance with the drawing, the specified performance characteristics shall govern.

3.3 Wire, hookup. - Conductor size for hookup wire shall be sufficient to avoid deterioration in electrical properties of the wire from overheating.

3.3.1 Bare. - Bare hookup wire shall not be used unless insulated wire is impracticable because of circuit characteristics or shortness of the wire run. Bare hookup wire shall be type S, soft or drawn and annealed, and coated, and shall conform to Specification MIL-W-3861.

3.3.2 Insulated. - Insulated hookup wire shall be stranded, where practicable, and shall conform to table I; however, when the application is not covered therein, the wire to be used shall be as approved by the contracting officer for the application concerned.

Table I. - Hookup wire, insulated

Temperature rating (maximum hot spot)	Voltage ratinc (volts ac)	Type	Specification
To 176° F (Note A)	300	LW	MIL-W-76
	(Note B)	HW	
	500	FX	
To 275° F	600	KF	MIL-W-12349

Note A: In sealed enclosures containing heat-dissipating parts such as electron tubes, power transformers, or resistors, type KF wire conforming to Specification MIL-W-12349 shall be used unless another type of hookup wire is approved by the contracting officer for the application.

Note B: 600 volts in sizes 14 to 6.

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the current version before use.

3.4 Brazing and soldering. -

3.4.1 Brazing. - Brazing of steel, copper, copper alloys, and nickel alloys shall conform to Specification MIL-B-7883.

3.4.2 Soldering

3.4.2.1 Solder. - Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571.

3.4.2.2 Flux and cleaning agents. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.2.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment, flux being spread over a larger area, or resulting damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.5 Cleaning. - Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.6 Finish protective. - Units shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware type items such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.2)

3.6.1 Electrical contacts. - Surfaces used for electrical contact shall be silver or silver-alloy except where special contact metals are used.

3.7 Marking. -

3.7.1 General. - Marking shall conform to Specification MIL-M-13231.

Source: https://assist.dla.mil -- Downloaded: 20260-17T15:02Z
Check the source to verify that this is the current version before use.

3.7.2 Visibility. - Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 Serial numbers. - The following article shall be serial numbered:

Code Recorder RD60( )/U

3.8 Securing of parts. -

3.8.1 General. - Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain components and parts securely when the equipment is subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangments, or permanent separation of parts will not occur when the equipment is subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of equipment performance below full specification requirements. Where practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.8.2 Securing of parts by threaded fasteners. - Securing of parts by threaded fasteners to hard material or soft material (aluminum, magnesium, zinc, plastic, etc.) except wood, shall conform to the following:

3.8.2.1 General. - Except as specified below, a flatwasher shall be used between any screwhead and soft material unless the shape of the head is such that the flatwasher is unnecessary.

3.8.2.2 Mounting of soft material to soft material. -

(a) Through-screw secured by a locknut. This method is preferred.

(b) Nylok screw as made by the Nylok Corporation, New York, New York, or equal, in bushing, nut, or insert. The bushing, nut, or insert shall be permanently secured to the structure.

(c) Screw in tapped hole with a screw-lock insert as made by the Heli-Coil Corporation, Danbury, Connecticut, or a Tap-Lok insert as made by the Groove Pin Corporation, Ridgefield, New Jersey, or equal. The manufacturer's installation instructions and recommended tools shall be employed.

(d) Screw in tapped hole or through-screw with plain nut, with insulating compound conforming to Specification MIL-I-17384 applied around the screwhead or around the screwhead and nut, as applicable. This method shall be used only for small screw-sizes and where one of the foregoing means is impracticable.

(e) Stud is tapped hole. One end of the stud and the tapped hole in which it is inserted shall be National Coarse thread; the other end shall be National fine thread. A wire screw-thread insert shall be used if required for thread reinforcement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

(f) Screw in tapped hole. This method shall be used only for holding labels or similar lightweight parts not used for structural purposes or critical applications, and, in particular, shall not be used when frequent disassembly may be necessary. Use of a flatwasher in this application is not mandatory.

3.8.2.3 Mounting of hard material to soft material. -

(a) As specified in 3.8.2.2

(b) Screw, with a split helical lockwasher under screwhead, in bushing, nut, or insert. The bushing, nut, or insert shall be permanently secured to the structure.

(c) Screw, with split helical lockwasher under screwhead, in tapped hole protected by a wire screw-thread insert as made by the Heli-Coil Corporation, or equal. The manufacturer's installation instructions and recommend tools shall be employed.

3.8.2.4 Mounting of soft material to hard material. -

(a) As specified in 3.8.2.2

(b) Nylok screw, or equal, in tapped hole

(c) Self-tapping screw. (See 3.9)

3.8.2.5 Mounting of hard material to hard material. -

(a) As specified in 3.8.2.2 through 3.8.2.4

(b) Screw in tapped hole, with split helical lockwasher under screwhead.

3.8.3 Brittle material. - Brittle castings, or parts made of ceramic or other brittle material, shall not be over-constrained by the means used to secure them. Not more than three points of contact shall be used between the brittle parts and the mounting surface, unless the mating faces are accurately machined. Mounting washers of suitable plastic, rubber, or soft copper having slight compressibility shall be provided to prevent local breakage or cracking of the parts. Lead washers shall not be used. Threaded holes in ceramic material shall not be employed for assembly or mounting of parts.

3.9 Self-tapping screws. - Thread-forming screws shall not be used. Thread cutting screws shall be used only if construction is improved thereby and only in places where the screws will not require loosening or removal during operation or maintenance of the equipment. In addition, their use shall be subject to approval by the contracting officer. Chips formed in through holes by thread cutting screws shall be removed.

3.10 Tropicalization. -

3.10.1 Treatment of assembled equipment. - Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the currentversion before use.

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.

(2) It does not incorporate any material listed in 3.10.3 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3)It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.10.2 Treatment of materials. -

3.10.2.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.10.2.2 Toxicity. - Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.10.2.3 Flexibility. - Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.10.2.4 Statement of treatment. - The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with Specification MIL-T-152, the above statement shall be incorporated in the statement of treatment required by that specification.

3.10.3 Materials not resistant to moisture and fungi. - The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated
(b) Cotton
(c) Cork
(d) Felt, hair or wool
(e) Jute
(f) Leather
(g) Linen
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

Source: https://assist.dla.mil -- Downloaded: 202640-17T15:02Z
Check the source to verify that this is the current version before use.

3.11 Welding. - Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. Wherever practicable, the welded joint shall conform to Specification MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary.

3.11.1 Cleaning. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.11.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum or magnesium.

3.12 CD 1341( ). - Cord CD 1341( ) shall consist of an 8 foot length of 2 conductor #18 AWG cordage. The cordage shall conform to Specification MIL-C-3432. Buna S compound shall be used for insulation and jacket material. Class HG shall be imployed. The outside diameter shall not exceed 9/32. It shall terminate at one end in a Hubbel Connector #7257, or equal and at the other end in a connector UP-120/M, or equal. Connector UP-120/M shall conform to MIL-C-3767/3. The cord assembly shall conform to all requirements of Specification MIL-C-3885. This cord shall be used to connect the power source to the Code Recorder RD60( )/U. (See 4.12).

3.13 Tubes, electron. - Electron tubes shall conform to Specification MIL-E-1 and shall be of the types given in Standard MIL-STD-200. Each equipment that uses electron tubes shall be furnished with the same tubes used when the equipment was inspected. Unless separate packaging of tubes is specified, they shall remain in the respective sockets occupied during inspection of the equipment. Unless otherwise approved by the contracting officer, specified performance of the equipment shall be obtained without selection of tubes. If the contractor finds it impossible to meet specified equipment performance without selection of tubes, he shall so inform the contracting officer and include:

(a) Evidence that use of selected tubes cannot be avoided by adjustment or minor design changes in the equipment.

(b) Detailed information on the narrower limits or special characteristics of selected tubes.

(c) Information on availability of selected tubes for maintenance of the equipment.

3.14 Interchangeability. - Corresponding units and replaceable assemblies, subassemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the currentversion before use.

items are used. (See 4.18) When dimensions, ratings, characteristics, etc. are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly, or part is to be considered replaceable, the contracting officer shall be consulted.

3.15 Electrical requirements. -

3.15.1 Keying. - The recording pen shall move from the front stop to the rear stop when ground and contact terminals on the terminal strip at the rear of the chassis are shorted, and shall return to the front stop when the short is removed. (See 4.6.1)

3.15.2 Sensitivity requirement. - The 1000-cycle signal voltage input required to move the pen from the front stop to the rear stop shall be not more than 0.75 volt RMS (Root Mean Squared), measured across the low impedance input terminals strip at the rear of the chassis. (See 4.6.2)

3.15.3 Line voltage. - The recorder shall be capable of operation from either 115 or 230 volts, 50-60 cycle a.c. power sources. The operation of the recorder shall not be affected by a 10 percent variation from the rated input voltage of the indicated power sources when any one such power source is connected to the unit. (See 4.6.3)

3.15.3.1 Power consumption. - The total power consumption from the line voltage source shall not exceed 140 watts and the power required for standby shall not exceed 85 watts. (See 4.6.3.1)

3.15.4 Low frequency response. - The recorder shall be connected for low frequency response test, see paragraph 4.6.4 of this specification. The recorder shall mark and space without buzzing when the signal level control is rotated back and forth between positions "4" and "6" of the dial.

3.15.5 High frequency response. - The recorder shall be connected for high frequency response test, see paragraph 4.6.5 of this specification. The recorder shall mark and space when the signal level control is rotated back and forth between positions "4" and "6" of the dial.

3.15.6 Limiting circuit. - The recorder shall be connected for high frequency response test, see paragraph 4.6.6 of this specification. There shall be no appreciable change in the wave form observed between the two settings.

3.15.7 Monitor circuit test. - The recorder shall be connected for high frequency response test, see paragraph 4.6.7 of this specification. The signals shall be clearly heard and no change in volume shall occur when the signal level control is rotated.

3.15.8 Filter action. - The recorder shall be connected for filter action test, see paragraph 4.6.8 of this specification. The traces obtained on the cathode ray oscilloscope shall be observed to determine the effectiveness of the filter circuit in reducing or eliminating the noise pulses of short duration.

3.16 Dielectric strength (equipment. - The test voltage shall be d-c or peak a-c of

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the current version before use.

the same value, and shall be applied for 10 seconds between wiring and adjacent non-conductors and ground. (See 4. 7)

3. 17 Input signal. - The recorder shall be capable of recording telegraph code signals of audio frequencies of 500 to 5000 cycles per second, when keyed at speeds up to and including 1000 words per minute. The equipment shall operate satisfactorily over the ranges of signal input voltages of from 0. 75 volt rms to 25 volts rms for low impedance input and 1. 5 volt rms to 50 volts rms for high impedance input. The low impedance input shall be suitable for operation from a 500 ohm line and the high impedance input from a 5000 ohm circuit. (See 4. 9)

3. 18 Operational - The recorder shall be connected for operational test, see paragraph 4. 8 of this specification. During any of these test, there shall be no tendency for the paper tape to roll off any pulley. At the conclusion of the foregoing tests the recordings on the paper tape shall be correct and readable. These test shall be performed after completion of the moisture and fungus-proofing treatment.

3. 19 Mechanical. -

3. 19. 1 Tape speed. - The equipment shall be equipped with a speed control to adjust the tape travel to the speed of the signals being recorded. The accuracy of the calibrated speed control, when tested in accordance with 4. 13. 1 observed by running a 10 foot length of paper tape through the drive roller with the control set, shall be as listed below:

Speed Control Setting	Time in seconds for 10 foot length of tape
70	7 to 10
35	15 to 19
5	114 to 126

3. 19. 2 Tape traction roller. - The tape traction roller properly centered on the tape drive roller shall cause the tape to be pulled evenly without slippage or tendency to roll off the tape drive roller. There shall be no indentations on the paper which may be caused by excessive pressure of the traction roller when tested in accordance with 4. 13. 2.

3. 19. 3 Friction drive. - The friction drive of the recorder shall have a smooth puller action. There shall be no scuffing of the neoprene ring of the driven wheel due to excessive pressure between it and the drive cone. (See 4. 13. 3)

3. 19. 4 Ease of adjustment. - All controls shall turn easily and smoothly without excessively motion, binding, scraping or cutting. All switches shall have a positive mechanical index at each position, operating so as to require no searching for contact by the operator. A small amount of lubricant (graphite or equivalent) shall be applied if it does not interfere with the operation.

3. 20 Service conditions. - The equipment shall meet the following service conditions:

3. 20. 1 Equipment operating. - After a warmup period of 1/2 hour, the equipment

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the currentversion before use.

shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) Temperature. - Any ambient temperature in the range of -40° F to +125° F.

(b) Elevation. - Any elevation up to 10,000 feet above sea level

(c) Relative humidity. - Any relative humidity up to 98 percent

(d) Orientation. - Any orientation up to 20 degrees from the normal operation orientation.

(e) Dust. - Dust as encountered during heavy traffic over dry clay roads.

3.20.2 Equipment nonoperating. - Except as stated in 3.20.2(b), the equipment shall comply with the operational requirements of 3.20.1 after subjection to any of the following non-operating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) Temperature - Continuous exposure for 72 hours at 160° F and 72 hours at -80° F, at any orientation.

(b) Elevation. - Any elevation up to 50,000 feet above sea level.

(c) Relative humidity. - Relative humidity up to 100 percent, including condensation caused by temperature changes. Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity the equipment shall meet the requirements of 3.20.1 except that the output may be 25 percent less than specified value.

(d) Orientation. - Indefinite storage in any orientation.

(e) Vibration, bounce, and shock. - See 3.21

3.21 Vibration, bounce, and shock. -

3.21.1 Vibration. - Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second.

3.21.2 Bounce and shock. - The equipment shall meet the requirements of table II.

Table II - Bounce and shock

Requirement	Inspection Paragraph	Performance after inspection
Bounce; light equipment	4.14	Full specification performance. No physical damage except surface abrasion.
Shock; bench-handling	4.16	Full specification performance. No physical damage.

(Cont'd)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

Table II - Bounce and shock

Requirement	Inspection Paragraph	Performance after inspection
Shock; drop	4.15	Equipment shall be operable. Minor mechanical damage is permissible. (Note A)

Note A: After this shock test, equipment performance shall conform to the following:

There shall be no degradation of performance

3.22 Preconditioning. - Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4. (Also see 4.3(d).)

3.23 Preproduction samples. - The contractor shall furnish preproduction samples for approval, as required by invitation for bids and contract. (See 6.1(e) (1))

3.24 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (see 6.1)

3.25 Workmanship. - Items shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.4.2.2 Flux and cleaning agents for soldering
3.4.2.3 Process for soldering
3.5 Cleaning of equipment
3.8.1 General requirements of securing of parts
3.11.1 Cleaning of welding
3.11.2 Process for welding

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection; responsibility and classification of. -

4.1.1 Contractor responsibility. - Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. Inspection records of the examinations and test shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (see 4.2)

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.19).

(1) Procurement inspection of items before preparation for delivery. (see 4.3)

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the current version before use.

(2) Procurement inspection of preparation for delivery (see 4.19 and section 5.)

4.2 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table III and the group A and group B inspection specified in tables IV and V respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling, (4) shock, ballistic or drop, and (5) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table III Preproduction inspection

Inspection For additional preproduction inspection see 4.2	Rqt. Par.	Test Par.
Temperature test	3.20.1(a)	4.10
Altitude test	3.20.1(b)	4.11
Bounce test	3.21.2	4.14
Shock test; drop	3.21.2	4.15
Humidity	3.20.1(c)	4.17
Vibration test	3.21.1	4.21

Table IV - Group A inspection

Inspection or Test	Rqt. Para	Insp. Para	AQL Major	AQL Minor
Visual and Mechanical				
Recorder Main Chassis		4.5	2.5 DPHu	10.0 DPHu
Final (Code Recorder RD-60( )/U		4.5	1.5 DPHu	6.5 DPHu
Cord CD-1341		4.5	1.0%	4.0%
Calibration of tape speed	3.19.1	4.13.1)		
Adjustment of friction drive	3.19.3	4.13.3)	2.5 DPHu	*
Adjustment of tape traction roller	3.19.2	4.13.2)	combined for the group	
Electrical				
Keying test	3.15.1	4.6.1)		
Sensitivity	3.15.2	4.6.2)		
Line voltage test	3.15.3	4.6.3)	2.5 DPHu	*
Low frequency response	3.15.4	4.6.4)	combined	
High frequency response	3.15.5	4.6.5)	for the	
Limiting circuit test	3.15.6	4.6.6)	group	
Monitor circuit test	3.15.7	4.6.7)		
Filter circuit test	3.15.8	4.6.8)		
Input voltage test	3.17	4.9)		
Cord CD-1341( )				
Continuity	3.12	4.12)	1.0%	*
Dielectric strength	3.12	4.12)	combined	
Insulation resistance	3.12	4.12)	for the group	
Dielectric test (Equipment)	3.16	4.7	1.0%	*
Operational test	3.18	4.8	1.0%	*

*No minor defects, all defects are considered major.

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the current version before use.

Table V Group B inspection

Inspection	Rqt. Par.	Test Par.
Bench-handling	3.21.2	4.16

4.3 Procurement inspection of items before preparation for delivery.- The minimum inspection performed by the contractor shall be as specified by (a) through (d) below--to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. (see 6.3) Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of group A and group B inspection as specified in 4.3.1 through 4.3.3.2.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that the responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when it is considered more practical to select separate samples for group B inspection.

(d) Each unit which will be subjected to group A and group B inspection shall be preconditioned after final assembly. (See 3.22)

4.3.1 Group A inspection.- This inspection (including sampling) shall conform to table IV and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.8) shall be last. (see 6.5)

4.3.2 Group B inspection. - This inspection (including sampling) shall conform to table V and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.4 and the requirements of Standard MIL-STD-105 for disposition of rejected product.

4.3.2.1 Group B sampling plans.- The group B sampling plans, as listed in table V, shall be as follows: The AQL shall be 6.5% and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection.

4.3.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Source: https://assist.dla.mil -- Downloaded: 202640-17T15:02Z
Check the source to verify that this is the current version before use.

4.3.2.3 Procedure in case of failure. - When an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.3.3 Reinspection of conforming group B sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B inspection. or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.4 Disposition of nonconforming product. - When defective sample units or failed inspection lots are submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4 Bounce preconditioning. - The unit shall be placed in its normal operation position on the table of the Package Tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches, as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.5 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.6 Electrical tests. -

4.6.1 Keying test. - The recorder shall be connected to an a-c power source of 115 volts, 50 to 60 cycles. The signal level and pen power controls shall be at "o" and the filter switch placed in the OUT position. The keying action of the recording shall conform to paragraph 3.15.1 of this specification.

4.6.2 Sensitivity test. - The recorder shall be tested for sensitivity as follows:

A 1000 cycle signal voltage source shall be connected to the signal input terminals of the recorder. The "Pen Power" and "Signal Level" controls shall be set to "10". Sensitivity shall conform to paragraph 3.15.2 of this specification.

4.6.3 Line voltage. - The recorder shall be tested using the same connections described under sensitivity test of 4.6.2 with a 50 to 60 cycle ac line voltage, 230 volts rms with a ten percent variation, connected to the recorder. The sensitivity of the recorder shall be checked at 230 volts minus 10% and at 230 volts plus 10%. (See 3.15.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

4.6.3.1 Power consumption - The power consumption shall be checked at 230 volts plus 10% RMS, 50-60 cycle AC line voltage input. The power consumption shall not exceed requirements of 3.15.3.1.

4.6.4 Low frequency response. - The recorder shall be connected to 115-volts, 50 to 60 cycles, a-c power source and tested for low frequency response by applying, across the input terminals (ground and tone) of the recorder, a 500 cycle signal of sufficient voltage to cause the pen to move from the front stop to the rear stop with the "Signal Level" control set "5". (see 3.15.4)

4.6.5 High frequency response. - The recorder shall be connected to a 115 volt, 50 to 60 cycles, a-c power source and tested for high frequency response by applying, across the input terminals (ground and tone) of the recorder, a 5000 cycle signal of sufficient voltage to cause the pen to move from the front stop to the rear stop with the Signal Level control set at "5". (see 3.15.5)

4.6.6 Limiting circuit test. - A 1000-cycle signal potential of 20 volts rms shall be connected, through a keying wheel adjusted to give 250 words per minute, to the high impedance input terminals of the recorder. With a cathode ray oscilloscope, the traces shall be observed on the recorder with a Signal Level setting between "3" and "5" and one at "10" respectively. (see 3.15.6)

4.6.7 Monitor circuit. - The monitor circuit of the recorder shall be tested using the same arrangement of equipment as used for 4.6.6. A pair of high impedance headphones or a speaker shall be plugged into the jack on the front panel of the recorder. (see 3.15.7)

4.6.8 Filter action. - The recorder shall be tested, using the same arrangement of equipment as used for 4.6.6 to determine the effectiveness of the filter circuit in reducing or eliminating the noise pulses of short duration. The filter switch shall be set at the in position. (see 3.15.8)

4.7 Dielectric. - The dielectric strength of all circuits on each equipment shall be tested to withstand 400 volts or two times the maximum peak operating voltage, whichever is greater. (see 3.16)

4.8 Operational. - The recorder shall be given an operational test for not less than two minutes each at 1000, 400, 250, 150 and 50 words per minute. The power used shall be 115 volts, 50-60 cycles, a.c. and the input signal shall be 20 volts rms a.c. at 2000 cps connected through a motor driven keying wheel. Under each condition, the Filter Switch shall be in the out position, the Signal Lever control at the minimum point where pen is actuated by the signal and Pen Power control set at the minimum point for legible recording. (see 3.18) Other adjustments shall be:

Wds per minute	Type Pen	Minimum Pen Swing
1000	Fine	3/32 inch----------High
400	Fine	3/32 inch----------High
250	Medium	1/8 inch----------Medium
150	Medium	1/8 inch---------- Medium
50	Wide	1/8 inch---------- Low

Source: https://assist.dla.mil -- Downloaded: 202640-17T15:02Z
Check the source to verify that this is the current version before use.

4.9 Input signal. - The recorder shall be given operational test recording telegraph code signals under the following input signal conditions: (see 3.17)

a. 50 words per minute, 500 cycle per second, .75 volt rms into the 500 ohm tone input terminals.

b. 50 words per minute, 500 cycles per second, 25 volts rms into the 500 ohm input terminals.

c. 1000 words per minute, 500 cycles per second, .75 volt rms into the 500 ohm tone input terminals.

d. 1000 words per minute, 500 cycles per second, 25 volts rms into the 500 ohm tone input terminals.

e. 50 words per minute, 5000 cycles per second, 1.5 volts rms into the 5000 ohm tone input terminals.

f. 1000 words per minute, 5000 cycles per second, 50 volts rms into the 5000 ohm tone input terminals.

g. The test of 4.9(a) through 4.9(g) shall be performed with a power input voltage of 115 volts, 50-60 cycle a.c. under each condition, the filter switch shall be in the out position, the signal level control at the minimum point where pen is actuated by the signal and pen power control set at the minimum point for legible recording. Other adjustments shall be:

Words per minute	Type Pen	Minimum Pen Swing
1000	Fine	3/32 inch----------High
50	Wide	1/8 inch----------Low

4.10 Temperature test for equipment. - The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. The equipment shall meet full specification requirements when tested at step 4, step 5, and step 8.

4.11 Altitude test. - The equipment shall be placed in an altitude chamber at normal conditions of temperature, pressure, and humidity, and performance measured. The pressure shall then be reduced to 20 inches of mercury and stabilized for ten minutes. The performance of the equipment shall again be measured. The pressure shall then be lowered to 3.2 $\pm$ 1 inches of mercury and the chamber maintained at this pressure for a 10 hour period. The pressure shall then be increased to 29.9 inches of mercury and the performance measured again. There shall be no degradation of performance in the second and third measurements from those established in the first test.

4.12 Cord CD-1341( ). - Continuity, Dielectric strength, and insulation resistance test shall conform to the requirements of MIL-C-3885.

4.13 Mechanical test. -

4.13.1 Tape speed. - The record shall be threaded with a 10 foot length of tape. The unit shall be connected to an a-c power source of 115 volts, 50 to 60 cycles and the a-c power switch turned to the ON position. The Standby-Operate switch shall be

Source: https://assist.dla.mil -- Downloaded: 20261O-17T15:02Z
Check the source to verify that this is the current version before use.

turned to operate. The calibrated speed control shall be tested for compliance with 3.19.1

4.13.2 Tape traction roller. - The tape traction roller, tension, with a length of paper tape threaded through the roller, shall be tested for proper pulling action in accordance with 3.19.2.

4.13.3 Friction drive. - The record operated as in 4.4.1 shall be tested for compliance with 3.19.3.

4.14 Bounce test. - The equipment shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 28 rpm ± 1 percent, for a total of 3 hours. During each 1/2-hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.15 Shock test; drop. - The equipment shall be subjected to the drop test in accordance with table VI. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or a rigid steel frame.

Table V - Drop test

Weight of equipment including case	Largest dimension of case (inches)	Notes	Height of drop (inches	Number of drops
100 to 200 pounds, inclusive	under 36	A	30	Drop on each corner. Total of 8 drops.

Note A: Drops shall be made from a quick-release hook; or pendulum tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal.

4.16 Shock test; bench-handling. - The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)" with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.17 Moisture-resistance test for equipment- The equipment shall be tested as follows:

4.17.1 Test conditions. -

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the current version before use.

(a) Do not remove equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.17.2 Test method. -

(a) Dry at 130° ± 5° F for 24 hours.

(b) Condition at 77 ° ± 5° F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.17.3 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for 5 58-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.17.3 shall be made during the periods shown on the standard and shall comply with 4.17.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.17.3.

4.17.3 Performance. - Measurements made during test, and specified performance during cycling, shall be as follows:

Paragraph	Measurements	Performance during cycling
4.6	Electrical test	Full specification requirement
4.8	Operational test	Full specification requirement

4.17.4 Failure. - If any equipment fails to meet the performance specified in 4.17.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.17.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.17.2(e), it does not pass the test.

4.18 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.14. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.19 Procurement inspection of preparation for delivery. - The minimum inspection performed by the contractor shall be as specified in 4.19.1 and 4.19.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government - at its discretion - will verify, by product inspection, the contractor's compliance with specified requirements. (see 6.3.) Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.19.1 Preservation and packaging. - Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table VII.

TABLE VII

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	2. Air not expelled from barrier.
3. Incorrect packaging method applied.	3. Conforming wraps are not snug-fitting and contain voids.
4. Cushioning or padding omitted.	4. Any item of marking information other than 8 thru 11 omitted, incorrect, or illegible.
5. Cushioning or padding inadequate for the protection of the barrier material from projections, sharp edges, or other similar features of the item.	5. Packaging material damaged.
6. Cushioning inadequate for the physical and mechanical protection of the item.	

Source: https://assist.dla.mil -- Downloaded: 202640-17T15:02Z
Check the source to verify that this is the currentversion before use.

TABLE VII - cont'd.

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR

7. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.

8. Stock number omitted, incorrect, or illegible.

9. Nomenclature omitted, incorrect, or illegible.

10. Marking of quantity of items in package omitted, incorrect or illegible.

11. Different stock numbered items in the same unit package.

4.19.2 Packing and marking of exterior containers. - Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table VIII.

TABLE VIII

PACKING, AND MARKING THEREOF

MAJOR

1. Use of improper or defective material.

2. Quantity in pack not as specified.

3. Box closure not as specified.

4. Type, grade, class, and style of the shipping container not as specified.

5. Strapping omitted.

6. Strapping inadequate or incorrectly applied.

MINOR

1. Unsealed carton.

2. Defective taping or sealing of carton.

3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.

4. Any item of required marking information other than 8 thru 14 listed under major defects omitted, incorrect, or illegible.

Source: https://assist.dla.mil -- Downloaded: 202640-17T15:02Z
Check the source to verify that this is the current version before use.

TABLE VIII - cont'd.

PACKING AND MARKING THEREOF

MAJOR

7. Shipping documents or packing list omitted.

8. Stock number omitted, incorrect, or illegible.

9. Nomenclature omitted, incorrect, or illegible.

10. Marking of quantity of items in pack omitted, incorrect, or illegible.

11. Destination marking omitted, incorrect, or illegible.

12. Service designation (color marking) omitted.

13. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.

14. Overseas code marking omitted, incorrect, or illegible.

4.19.2.1 Inspection lot. - A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.19.2.2 Procedure in case of failure. - If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.19.2.3 Disposition of nonconforming product. - Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the currentversion before use.

4.20 Rough handling test (preparation for delivery). - When rough handling test is required by the contract (see 6.1(e)). The following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling.

Paragraph	Measurement:	Performance:
4.13	Mechanical test	Full specification requirement
4.6	Electrical test	Full specification requirement
4.8	Operational test	Full specification requirement

4.21 Vibration test. - The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.21.1 as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. The vibration table shall provide approximately sinusoidal vibration. (See 4.21(d).)

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because the unit weighs more than 250 pounds or because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. - Recorder, Code, Tape (Code Recorder RD-60( )/U) shall be preserved, packaged, and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described herein.

5.1.1.1 Technical manual. - Each technical manual shall be packaged Method IC-3 as follows: Inclose each technical manual within a close-fitting type I, class b bag, conforming to Specification MIL-B-117. Heat seal the bag to effect closure.

Source: https://assist.dla.mil -- Downloaded: 202640-17T15:02Z
Check the source to verify that this is the current version before use.

5.1.1.2 Tubes. - All tubes over four inches in length, not securely clamped in place, shall be removed and together with spare tubes, shall be packaged individually in accordance with the applicable provisions of Specification MIL-P-75.

5.1.1.3 Cord CD-1341, Sleeve assembly (spare). - Each item shall be packaged Method IC-3 as follows: Wind the cord into a coil of proportionate dimensions and tie at a minimum of 3 places with 6 ply cotton wrapping twine, type II, conforming to Specification T-T-871, securing the ends of the cord to the coil with the ties. Cushion the coiled cord and the sleeve assembly individually with type I, single-face, flexible corrugated fiberboard conforming to Specification LLL-F-291. Secure cushioning with class 1, gummed paper tape, conforming to Specification UU-T-111. Continue packaging as specified in 5.1.1.1.

5.1.1.4 Disassembly. - Remove the inkwell from the face of the recorder by loosening the thumb screw which holds the inkwell on its mounting bracket, and remove the inkwell by sliding it out of the slot in the bracket. Tighten the thumb screw securely.

5.1.1.5 Inkwell. - Each inkwell shall be packaged Method III as follows: Cushion each inkwell by wrapping in single-face flexible corrugated fiberboard. Secure cushioning with gummed paper tape. Place the cushioned inkwell within a close-fitting style II, Type A, class 2 folding carton conforming to Specification PPP-B-566. Closure shall be in conformance with the specification.

5.1.1.6 Consolidated package. - Consolidate the items packaged as specified in 5.1.1.2 through 5.1.1.5 together within a close-fitting RS style CF box, conforming to Specification PPP-B-636, class 1. Close the box in accordance with the appendix of the box specification.

5.1.1.7 Stowage of spare parts and tools. - Stow and secure spare parts and special tools within the kit attached to the tape bin on the face of the controller. Fill all voids with single face flexible corrugated paper. Close the kit and secure all fastenings.

5.1.1.8 Code Recorder RD-60( )/U. - Each recorder shall be packaged Method IC-5 as follows: Secure the tape drive mechanism in the recorder from movement with the shipping clamp provided. Position and secure the lid of the case. Place over the face of the recorder a blocking assembly fabricated of a wood panel and wood blocks to bridge projections on the face of the recorder. Surfaces of the blocks contacting the face of the recorder shall be covered with a minimum of 2 thickness of single-face flexible corrugated fiberboard, secured to each block with gummed paper tape. Cushion each recorder on all surfaces with cells or pads or both fabricated of double faced corrugated fiberboard having a minimum dry bursting strength of 200 pounds and designed to absorb the shock of impact encountered in handling and transit. Place the cushioned recorder and consolidated package (see 5.1.1.6) within a close-fitting RS style, W5c, CF box conforming to Specification PPP-B-636, class 2. Box closure shall be in accordance with the box specification and in addition seal all seams and joints with pressure sensitive water resistant tape, conforming to Specification PPP-T-76.

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:02Z
Check the source to verify that this is the current version before use.

5.1.2 Level C.- Code Recorder RD-60( )/U shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.-Code Recorder RD-60( )/U shall be packed for shipment one each, within a close-fitting, style 4, class 2, nailed wood box, conforming to Specification PPP-B-621. Place the technical manual (see 5.1.1.1) between the contents and lid of the box. Box closure shall be in conformance with the applicable box specification.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.2.2 Level B.- Code Recorder RD-60( )/U shall be packed as specified in 5.2.1 except the box shall be of domestic style or class as applicable.

5.2.3 Level C.- Code Recorder RD-60( )/U shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4 and Consolidated Freight Classification, Rules 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking.-Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.
(b) Type required.
(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)
(d) The specific paragraphs of section 5 which are applicable to the particular procurement.
(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

performance measurements can be made on the second sample. (see 3.23.)

(2) Source of color chips (see 6.2.)

(3) Preproduction packs as follows:

a. Makeup of packs.

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction packs and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction packs.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required. (3.24)

(i) Quantity of tools and running spare parts required. (see 3.24)

(j) Submission of the statement of treatment referenced in 3.10.2.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.2 Color. - The color chip furnished by the procuring agency will match color chip No. 2430 of superseded Specification TT-C-595. No matching color chip has been provided in the superseding Federal Standard No. 595. (see 3.6.)

6.3 Verification inspection. - The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A -- The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B -- That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

6.4 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: RD-60W/U. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information. (See 1.1.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

6.5 Location of operational inspection. - It is desirable that the operational inspection (4.8) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Definitions. -

6.6.1 Inspection terms. - The inspection terms used herein conform to Standard MIL-STD-109.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.